Training App — User Audit

Simulated first-time-user walkthrough & capability audit

> **Build audited:** commit `6290d46` , Android debug build on emulator (API 35, GMT)
> **Date:** 28 August 2026
> **Method:** new-user task walkthrough on device, cross-checked against the codebase for capabilities that exist in the data layer but are unreachable from the UI
> **Baseline:** 179 automated tests passing, `flutter analyze` clean

This audit simulates a person picking the app up for the first time and trying to do the obvious things: find an exercise, build a plan, train, and look back at what they did. Findings are ordered by how likely they are to block or mislead a real user, not by how hard they are to fix.

Summary of findings

#	Severity	Finding	Area
1	High	A plan cannot be deleted or renamed	Plans
2	High	A completed workout cannot be reviewed afterwards	History
3	Medium	Calendar says “No sessions on this day” before any day is picked	Calendar
4	Medium	Leaving a session mid-workout silently discards it	Execution
5	Medium	A library exercise cannot be edited after creation	Exercise library
6	Medium	Progression advice always reads in kg, ignoring the lbs setting	Units
7	Low	Body measurements are cm-only with no imperial option	Units
8	Low	Imported sessions show a placeholder day (“Any”)	Plans
9	Low	No search or filter inside a large plan	Plans
10	Fixed	Chart Y-axis labels overlapped near the top	Progress
11	Fixed	Per-exercise history was unreachable from the dashboard	Progress
12	Fixed	History chart axis had no unit label	Progress

Detailed findings

1. A plan cannot be deleted or renamed High

What happens: After a few imports the plan list accumulates entries — during this audit the device held *"training_plan_template (second edit)"* and *"neu"* alongside the real programme — with no way to remove or rename any of them.

Why it matters: The only route to a clean list is *Settings → Delete all data*, which also destroys every logged workout. A user tidying up their plan list can lose their entire training history in the process.

Note: `PlanDao.deletePlan()` already exists and is unit-tested, including correct cascade behaviour. This is purely a missing UI entry point, so it is a small change with a large usability payoff.

2. A completed workout cannot be reviewed afterwards High

What happens: Finishing a session writes a full record — every set, rep, weight, RPE and PR flag. None of it is readable afterwards. The only views over history are per-exercise charts and the Progress dashboard.

Why it matters: "What did I actually do on Monday?" is one of the most common questions a trainee asks, and it is the natural way to decide what to load next session. The data is captured but effectively write-only.

Suggested shape: Tapping a completed (green-check) day on the calendar opens that session's log. `SessionLogDao.getLogsForPlanSession()` and `getLoggedSetsForSessionLog()` already return everything needed.

3. Calendar shows a misleading empty state Medium

What happens: On opening the Calendar tab, the panel below reads *"No sessions on this day."* — while today's date visibly carries a scheduled-session dot.

Cause: `_selectedDay` starts as `null`, so the list is empty until the user taps a date, but the message is phrased as though a day had been chosen and found empty.

Fix: Either default the selection to today, or word the no-selection state differently ("Select a day to see its sessions"). The first is likely better — today is almost always what the user wants.

4. Leaving a session mid-workout silently discards it Medium

What happens: Execution mode holds all logged sets in memory and only writes to the database on *Finish session*. Pressing back, or being pulled away by the OS, discards everything with no warning and no recovery.

Why it matters: This was an accepted simplification early on, but the app now actively invites leaving mid-session — the floating rest timer exists precisely so users can switch tabs during a rest. The feature encourages the exact behaviour that loses data.

Fix (cheap): A confirmation prompt on back-navigation when any set is logged. **Fix (better):** Persist sets as they are logged and treat *Finish* as closing the session.

5. A library exercise cannot be edited after creation Medium

Creating a custom exercise captures the full schema, but nothing can be corrected afterwards — a typo in the name, or a missing video link, is permanent. There is also no way to delete a custom exercise, although `ExerciseDao.deleteExercise()` exists. The detail screen is read-only.

6. Progression advice ignores the lbs setting Medium

With the weight unit set to pounds, the auto-progression suggestion on the history screen still reads in kilograms — for example *"try 62.5kg next"* — because `progression.dart`'s `formatWeight()` hard-codes the suffix and does no conversion. Every other weight display converts correctly.

This is a correctness bug, not just cosmetic: a user training in pounds is being handed a number in the wrong unit, with the wrong unit named next to it.

7. Body measurements are centimetre-only Low

Circumference fields are labelled and stored in centimetres regardless of the weight-unit preference. A user who selected pounds has reasonably signalled a preference for imperial units, and will likely expect inches. Bodyweight itself does convert correctly.

8. Imported sessions show a placeholder day Low

Sessions imported without a weekday display as *"Week 1 • Any • Block 1 – Legs"*. "Any" is an internal placeholder surfacing in the UI; it reads as though it were a real choice the user made. Omitting the segment entirely would be clearer.

9. No search inside a large plan Low

The nine-block programme used for testing contains 30+ sessions in a single scrollable list. There is no search, filter, or collapse-by-block. The exercise library has search; plan detail does not, and it is the screen where a long list is most likely.

Fixed during this audit

10–12. Chart and labelling defects Fixed

- **Y-axis label overlap.** The volume chart drew "1K" and "1.1K" on top of each other. Fixed by a shared `niceAxis()` helper using Heckbert "nice numbers" stepping: the axis maximum is now always an exact multiple of the gridline interval, so the top value *is* a tick and no stray label can crowd one. Applied to the history chart as well, replacing its hard-coded version.
- **Dashboard rows were dead ends.** The Progress dashboard listed each exercise's latest top set but could not be tapped, so the trend behind that number was unreachable. Rows now open the exercise's full weight-over-time chart.
- **Unlabelled chart axis.** The history chart's Y axis showed bare numbers with no unit. It now names the active unit in the title.

What held up well

- **Data safety.** Backup/restore round-trips cleanly, including forward-compatibility: a backup taken before body measurements existed still restores. Destructive actions are consistently confirmed.
- **Unit handling.** Weight is stored in kilograms throughout and converted only at display boundaries — the correct architecture, and it holds everywhere except the progression text noted above.
- **Notifications.** The Android 13 permission prompt fires at the right moment, the switch correctly stays off if permission is refused, and scheduled alarms were verified landing at the right wall-clock times via `dumpsys alarm`.
- **Accessibility.** Colour is never the sole carrier of meaning — muscle areas always show their name as text beside the colour — and the palette picker previews real swatches so the choice can be made by eye.
- **Empty states.** Newly added screens explain what to do rather than showing a blank page. The calendar is the one exception, noted above.

Recommended order of work

1. Delete/rename a plan (#1) — highest ratio of user pain to implementation cost; the data layer is already done and tested.
2. Session log review (#2) — unlocks data the app already collects.
3. Progression unit bug (#6) — small, and currently shows wrong numbers.
4. Abandoned-session protection (#4) — at minimum a confirmation prompt.
5. Calendar empty state (#3) — a one-line change.
6. Edit/delete a library exercise (#5).
7. Remaining low-severity items (#7–#9) as convenient.

Training App — local-first training planner. Audit performed against commit `6290d46`. Findings 10–12 were fixed in the same pass and are recorded for completeness.